

Computational Identification of Unknown Organic Contaminants and Molecules in Remote Environments via High-Resolution Mass Spectrometry

Raelyn Brooks

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1 Goal

This project aims to characterize the chemical footprint of global pollution in remote high-altitude environments by identifying unknown organic contaminants in snowmelt and river water using high-resolution mass spectrometry (HRMS). I hypothesize that long-range atmospheric transport leads to the accumulation of diverse and previously uncharacterized organic molecules in these regions. By combining non-targeted analysis with molecular networking, this study will reveal both the identity and distribution patterns of these contaminants.

2 Specific Aims

2.1 Aim 1: Identify and characterize unknown organic contaminants in high-altitude water samples

- **What (Aim):** Use HRMS-based non-targeted analysis to detect and identify unknown organic molecules in snowmelt and river water collected from high-alpine regions.
- **How (Approach):** Water samples will be analyzed using liquid chromatography–HRMS, followed by computational workflows including molecular networking and suspect screening to annotate molecular features and cluster related compounds.
- **Impact:** This aim will generate a comprehensive chemical profile of contaminants in remote environments, moving beyond traditional targeted methods and uncovering “known

unknowns” and novel compounds.

2.2 Aim 2: Quantify spatial patterns and concentration gradients of detected contaminants

- **What (Aim):** Determine the concentration and distribution of identified compounds across different sampling sites and environmental matrices.
- **How (Approach):** Relative or semi-quantitative analysis will be performed using HRMS signal intensities, with comparisons across locations.
- **Impact:** This aim will establish how contaminant levels vary geographically, providing insight into deposition patterns and environmental accumulation in high-altitude systems.

2.3 Aim 3: Assess long-range atmospheric transport of contaminants

- **What (Aim):** Evaluate the relationship between detected contaminants and atmospheric transport processes to determine how far pollutants travel from their sources.
- **How (Approach):** Chemical profiles will be compared with known atmospheric markers and literature data; patterns in molecular classes and volatility will be used to infer transport mechanisms (e.g., long-range atmospheric transport and cold trapping).
- **Impact:** This aim will link chemical presence to global transport pathways, improving understanding of how human-generated pollutants reach remote ecosystems.